Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q02] | Part A (Q1-Q5)

Answer: 243

1. Each square strip is crossed exactly once, using its bottom edge, top edge, or rising diagonal. These are three independent choices per strip.

2. The ten choices determine all vertical moves: at each vertical line use its edge if and only if arrival and departure heights differ, including the starting and ending lines. This never repeats a segment and yields a bijection with the 3^{10} choice strings.

3. Thus $\sqrt{N} = 3^5 = 243$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2005-II-Q06] | Part B (Q6-Q10)

Answer: 392

1. A card $i \leq n$ moves to top-based position $2(n-i)+1$, while $i > n$ moves to $2(2n-i)+2$. Thus an odd fixed position must belong to A.

2. For $i = 131$, solve $131 = 2(n-131)+1$, giving $n = 196$. Card 262 from B is also fixed because $2(392-262)+2 = 262$. Thus the required total is 392.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2006-II-Q08] | Part B (Q6-Q10)

Answer: 336

1. The central triangle stays fixed under every symmetry, while the three corner triangles may undergo all six permutations. Hence only the center color and multiset of corner colors matter.

2. Three-color multisets selected with repetition from six colors number $\binom{8}{3} = 56$. Multiply by six center choices to obtain 336.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-I-Q10] | Part B (Q6-Q10)

Answer: 202

1. Write $a_i = 10b_i + c_i$ with digits b_i, c_i. Then the equation is $10B + C = 2010$, where $B = \sum b_i 10^i, C = \sum c_i 10^i$.

2. Conversely, every choice $B = 0, \ldots, 201$ gives nonnegative $C = 2010 - 10B$. Both are at most four-digit numbers, so padding their decimal digits with zeros uniquely recovers the coefficients.

3. This is a bijection with 202 choices of B, so the count is 202.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-II-Q10] | Part B (Q6-Q10)

Answer: 163

1. Write $f(x) = a(x-r)(x-s)$, so $ars = 2010 = 2 \cdot 3 \cdot 5 \cdot 67$. Allocate each prime to one of $|a|, |r|, |s|$: there are $3^4 = 81$ ordered magnitude triples.

2. For each allocation, four sign patterns have positive product. There are therefore 324 signed ordered triples (a, r, s). Swapping r, s identifies polynomials.

3. Only $r = s = 1$ and $r = s = -1$ are fixed under the swap, both with $a = 2010$. The number of orbits is $(324 + 2)/2 = 163$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2011-II-Q06] | Part B (Q6-Q10)

Answer: 080

1. The reflection $x \mapsto 11 - x$ exchanges $d - c > b - a$ and $d - c < b - a$. Thus the desired count is one half of the difference between $\binom{10}{4}$ and the equality count.

2. Equality has outer gaps $k = b - a = d - c$ and middle gap $j = c - b$. For $k = 1, 2, 3, 4$, positive integers (a, j) satisfy $a + j \leq 10 - 2k$. Their counts are $\binom{8}{2}, \binom{6}{2}, \binom{4}{2}, \binom{2}{2}$, totaling 50.

3. Therefore the answer is $(210 - 50)/2 = 80$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2018-I-Q10] | Part B (Q6-Q10)

Answer: 004

1. Code counterclockwise or inward as X, clockwise or outward as Y. From any state there is exactly one move of each type, so binary words describe paths bijectively. A step ends on the inner ring exactly when its code is X.

2. If the final code is X, spoke moves balance inward and outward, so the net angular displacement equals $\#X - \#Y$ modulo five. A return therefore requires $2\#X - 15 \equiv 0 \pmod 5$, with $\#X = 5, 10, 15$.

3. Fixing the last X, the count is $\binom{14}{4} + \binom{14}{9} + \binom{14}{14} = 1001 + 2002 + 1 = 3004$, giving remainder 4.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2023-II-Q10] | Part B (Q6-Q10)

Answer: 144

1. First fill with residues $0, 1, 2$, each appearing four times; then permute the four actual integers within each class, contributing $(4!)^3$. Adjacent residues must differ.

2. For a column (a, b) with remaining residue c, the next column has exactly the three possibilities $(b, a), (b, c), (c, a)$. Thus each choice of its unordered residue pair uniquely determines the orientation once the previous column is known.

3. For each residue to occur four times in six columns, the three unordered pair types must each occur twice. There are $6!/(2!)^3 = 90$ pair sequences and two orientations of the first column, yielding 180 residue fillings.

4. Hence $N = 180 \cdot 24^3 = 2^{11}3^5 \cdot 5$. Its divisor count is $(11 + 1)(5 + 1)(1 + 1) = 144$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2024-II-Q09] | Part B (Q6-Q10)

Answer: 902

1. A maximal placement has no empty row or column: an empty row allows a chip matching some column color to be added. Assign each row and column its occupied color.

2. A cell is filled exactly when its row and column colors agree; otherwise it must remain empty. Thus rows and columns must use the same set of colors. This gives a bijection with pairs of row/column colorings using either one common color or both colors.

3. There are two monochromatic possibilities and $(2^5 - 2)^2$ mixed possibilities. Total $2 + 30^2 = 902$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q13] | Part C (Q11-Q15)

Answer: 107

1. A three-element consecutive block makes any pairing impossible. Every maximal consecutive block therefore has size one or two. A two-block is forced to map outward; a singleton can move one unit left or right. Only blocks separated by one missing integer can compete for a target, so group them into chains with such gaps.

2. For fixed disjoint S, T, the pairing is unique if it exists: the allowed edges are edges of the integer line, whose finite components are paths, and an even path has only one perfect matching. Thus counting legal shifts counts image sets without multiplicity.

3. A chain of k singleton blocks permits $k+1$ images: an initial run shifts left and the remainder shifts right, since a right-then-left pattern collides. A chain with one two-block has one image; with two two-blocks it has none. Different chains are independent, so positive counts factor as $\prod(k_i+1)$; two-blocks cannot help minimize size.

4. To minimize $\sum k_i$ for product $4040 = 2^3 \cdot 5 \cdot 101$, split every composite factor gh into g, h, since $(g-1)+(h-1) < gh-1$. Cost $1+1+1+4+100 = 107$, realized by separated alternating-integer chains of those five lengths.

Source / 来源